Sentinel Eyes: Intelligent Sensor Network For Smarter Boarder Security

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Abstract

Border areas are subjected to continued threats from illegal border crossing, smuggling, and unauthorised movement. Many of the traditional methods used to monitor borders (such as manual patrol and fixed camera systems) have significant limitations including poor visibility, adverse weather conditions, large workforce needs, and high false alarm rates. The researchers have developed a novel solution called "Sentinel Eyes" which is an intelligent multi-sensor system powered by Al/ML algorithms. The Al/ML algorithms use data collected from a range of sensors (PIR Motion Detectors; Ultrasonic Distance Sensors; Magnetic Tamper Sensors; Camera Modules) to improve the accuracy of the classification of events into categories of either human, animal, vehicle, or natural disturbances, and therefore reduce false alarm rates and increase the speed and efficiency of responses. The Sentinel Eyes System has been tested in a real time environment demonstrating successful intrusion detection, prompt alert generation, and visualisation of alerts on a dashboard. It is considered a suitable example of a practical, cost effective, and scalable framework for the current challenges associated with border security.

The problem of border surveillance remains one of the most important national security issues of today. One of the major reasons for the problem of border surveillance is the size of many of the border areas, along with the harsh weather conditions, limited visibility and the need for large workforces to continuously monitor them. In addition, many of the traditional systems currently being used to monitor borders (i.e., manual patrol and fixed camera systems) suffer from two primary problems. Firstly, they do not provide adequate surveillance coverage over the entire border area, due to their inability to cover long distances with adequate resolution. Secondly, when they do detect something unusual they can also produce false alarms due to environmental factors (e.g., wildlife) or technical faults. To address these problems the researchers have designed and constructed a new type of border surveillance system called "Sentinel Eyes". The system uses four types of sensors (PIR Motion Detectors; Ultrasonic Distance Sensors; Magnetic Tamper Sensors; Camera Modules), and lightweight Al/ML algorithms to perform real time sensor integration and event classification to improve the detection of human presence on the border. The processing occurs at the edge, allowing rapid response times, while also conserving power consumption and bandwidth usage. In addition, the prototype demonstrates reliable intrusion detection, prompt alert generation, and visualisation of alerts on a dashboard with low false alarm rates. The low cost, scalability, and robustness of the Sentinel Eyes System make it a viable and efficient solution for monitoring borders, protecting military zones, and providing remote surveillance in developing countries.

Keywords

Border Security, Intelligent Sensor Network, IoT Surveillance, Machine Learning, Intrusion Detection, PIR Sensor, Ultrasonic Sensor, Magnetic Sensor, Real-Time Monitoring, Sensor Fusion, False Alarm Reduction, Edge Computing, AI-Based Security Systems, Smart Surveillance, Sentinel Eyes.

Introduction

Monitoring large and remote border areas for a country's safety and sovereignty is an extremely difficult job, largely due to harsh terrain, poor weather conditions, limited sight and the need for constant human presence. Manual patrols and CCTV cameras do not provide sufficient security because they are expensive to man and provide limited cover; also both fail in very low light and fog.

Additionally, many of these systems create false alarms from animals, wind or other environmental noise causing wasteful use of resources and delays in responding to true threats.

Advancements in IoT, sensor networks, and AI have opened up new opportunities to create intelligent, autonomous surveillance systems which will enhance awareness and reduce the amount of work required by people.

Smart sensor technology can be used to gather information in real time and AI/ML can then analyze the data to find meaningful patterns allowing for rapid identification of intrusion.

This study proposes the development of a multi-sensor AI driven surveillance system named Sentinel Eyes. The system will be a cost-effective, lightweight and mobile solution that will combine PIR motion sensors, ultrasonic sensors, tamper magnets, and camera sensors to continue to monitor and track activity in border areas and detect anomalies.

A lightweight machine learning model will process sensor data in real time to differentiate between humans, animals, vehicles and false alarms. Once an event has been identified the system will automatically send out alerts, log the event and provide visual confirmation of what was seen through a central dashboard.

Sentinel Eyes utilizes sensor fusion, real-time analysis and AI based classification to provide an effective and efficient method to improve border security, reduce false alarms and minimize the reliance upon manual review. This study will demonstrate how intelligent automation can greatly improve the performance and reliability of surveillance operations in resource constrained and sensitive environments.

Problem Statement

Border areas have a heightened need for ongoing monitoring due to potential risks from unauthorized cross-border movement (illegal crossing), smuggling, infiltration, etc. Current border surveillance systems however, primarily utilize manual patrols and static CCTV cameras with several constraints. Manually patrolling borders requires significant manpower resources, is very labor intensive and does not allow 24/7 coverage over large or remote terrain. In addition, static CCTV cameras provide limited viewing capabilities and fail in poor weather conditions; such as fog, heavy rain, dust, or at night.

One of the biggest issues with today's border surveillance systems is their high false alarm rate caused by wild animals, wind, or uncontrolled natural environment events. The excessive number of non-productive false alarms that these systems generate lead to wasted personnel time, delayed responses, and decreased confidence in the overall effectiveness of the monitoring systems. Additionally, today's border surveillance systems do not possess the intelligent analytical tools necessary to automatically differentiate between normal activity and suspicious activity.

Dataset Overview

This system uses four major types of data:

1. Motion Data (PIR Sensor)
2. Distance Data (Ultrasonic Sensor)
3. Magnetic Tamper Data (Reed / Magnetometer)
4. Image Frames (Camera Module)
5. Optional Environmental Data (Vibration, Temperature, Sound)

The dataset below is designed to help your AI/ML model classify events as:

- Human
- Animal
- Vehicle
- False Alarm (Wind/Noise)
- Fence Tampering

2. Sample Dataset Structure (Tabular)

This is how your CSV/Excel dataset should look:

- Timestamp PIR
- Motion

- Distance(cm)
- Sound
- Vibration
- Magnetic

	Time	Motion	Distance	Magnetic	Sound	Vibration	Status
1	2025-11-11 00:05:43	0	1.52	1	30.52	0	Safe
2	2025-11-11 00:05:45	0	9.1	1	41.24	0	Safe
3	2025-11-11 00:05:46	0	6.22	1	32.25	1	Safe
4	2025-11-11 00:05:47	1	6.85	0	41.69	1	Safe
5	2025-11-11 00:05:48	1	5.94	0	75.36	0	Safe
6	2025-11-11 00:05:49	1	9.37	0	88.14	0	Safe
7	2025-11-11 00:05:51	0	6.33	0	61.46	1	Safe
8	2025-11-11 00:05:52	1	8.01	1	80.12	1	Safe
9	2025-11-11 00:05:53	1	3.78	0	84.62	0	INTRUSION DECTECTED
10	2025-11-11 00:05:54	1	5.0	0	89.63	0	Safe

3. Explanation of Each Feature

1. PIR Motion (0 or 1)

- 0 = No motion
- 1 = Motion detected (human/animal/vehicle)

2. Distance (from Ultrasonic Sensor)

- Sudden changes indicate movement
- Very low distance <70 cm = vehicle or human close
- Medium 70–150 cm = walking human/animal

3. Magnetic Status

- Normal = No tampering
- Tampered = Fence or gate disturbed

4. Vibration Level

- Low / Medium / High
- Helps detect running vehicle, jumping animal, or tampering

5. Label (for training ML)

- Human

- Animal
- Vehicle
- False Alarm
- Fence Tampering

4. Folder Structure for Dataset

Use the following folder layout:

- Sentinel_Eyes_Dataset/
- |
- | ├── human/
- | ├── animal/
- | ├── vehicle/
- | ├── false_alarm/
- | └── tampering/
- ├── sensor_data.csv
- └── labels.csv

5. Minimum Dataset Size for Model Training

For a small ML model:

- Class
- Human
- Animal
- Vehicle
- False Alarm
- Recommended Samples
- 500–800
- 400–600
- 300–500
- 300–500
- Fence Tampering 200–300

Total recommended dataset = 1,500 – 2,500 samples

6. How to Collect This Dataset in Real Life

we don't need a real border — we can simulate:

Human Movement

- Multiple people walking at different speeds
- capture PIR + distance
- Animal Movement
- dogs, goats, birds, cats
- label as "animal"
- Vehicle
- cars, bicycles, bicycles
- from 2-5 meters away
- False Alarm
- wind moving trees
- no motion
- light vibration
- distance remains the same
- PIR = 0
- label = "false alarm"

- Tampering of Fence
- tap on a metal surface close to a magnetic sensor

Methodology

Sentinel Eyes utilizes an integrated, multi-tiered approach to utilize sensor hardware, IoT communications, data processing, artificial intelligence (AI)/machine learning (ML), and alerts to detect intruders. The overall design provides for real-time intrusion detection with minimal false positives. Below describes each tier of the overall methodology.

System Architecture Design

The system was designed utilizing a decentralized sensor network (or mesh) that spans a border area. All nodes consist of:

• PIR motion sensor

• ultrasonic distance sensor

• magnetic tamper sensor

• camera module

• microprocessor/Raspberry Pi

A centralized processor node connects all sensors together for data collection and processing purposes.

Sensor Placement and Calibration

Each sensor was strategically placed to create maximum coverage:

PIR Sensor Calibration

• detects movement from a thermal source

• calibrated to detect motion at distances of 3-7 meters

Ultrasonic Sensor Calibration

• measures the distance between a moving object and the sensor

• threshold levels set to measure distance change of >10cm

Magnetic Sensor Placement

• mounted on gates or fencing

• can detect tampering or forced entry into the fence/gate

Camera Module Placement

• captures low resolution images when events trigger

• used to support the ML Model

Data Collection and Data Processing

Real time sensor data is collected as follows:

• Motion reading (0/1)

• Distance readout in cm

• Status of magnetic sensor (Normal/Tampered)

• Vibration/Environmental Values

Prior to the data being utilized for machine learning, preprocessing takes place to remove noise, normalize distance values, synchronize time across the various sensors, and label the data as human/animal/vehicle/false alarm. This labeled data will be used to train the ML model.

Data Transmission and IoT Communications

Each sensor node sends data to the processor node via:

• Wi-Fi/MQTT protocol

• local server or raspberry pi

• edge computing to eliminate delay in transmission

All events are sent to the processor in real time for evaluation.

Developing a Lightweight Machine Learning Model

A lightweight machine learning model is developed to provide real-time classification of events using the sensor and image data collected. The steps include:

1. Feature Extraction

Features are extracted from:

Sensor Data

- PIR Status
- Distance Change
- Magnetic Field Change
- Vibration Pattern
- Time of Day

Image Data

- Basic Shape/Motion Patterns
- Size Approximation of Objects

2. Model Selection

- Lightweight models that run on Edge Devices are selected for this project:
- Decision Trees
- Random Forest

- SVM
- Logistic Regression
- Lightweight CNN (for Image Support)

3. Training & Validation

- Data Set Split: 70% for Training & 30% for Testing
- Models are assessed based upon their accuracy, precision, and false alarm rates.
- The best performing model is then integrated into the overall system design.

4. Event Classification

Each event classified by the ML model is categorized as follows:

- Human Intrusion
- Animal Movement
- Vehicle Movement
- False Alarm (due to wind/noise/random activity)
- Fence Tampering

Classification enables the system to determine whether the activity being monitored is a true threat or simply harmless.

5. Real-Time Alert Generation

When an actual intrusion occurs, the system will generate an alert, record the date/time along with the sensor data and capture any image(s), send alerts via dash board, SMS, or Email, store the log of all events for later review.

This enables the system to rapidly respond to the threat.

6. Dashboard Monitoring System

The live monitoring system has a web-based or local interface for displaying the:

- Live sensor values
- Alerts generated
- Intrusion Log
- Captured Images
- Health of the Sensors
- Security personnel may view the current state of the area in real-time.

7. Testing and Performance Evaluation

The system is tested against a number of different test cases:

- Human Walking/Running
- Dog/Animal Movements
- Wind Shaking Objects
- Vehicle Movement
- Fence Tampering

Performance is measured as follows:

- Accuracy
- False Positive Rate
- Detection Speed
- Power Consumption

Results indicate improvements in performance compared to the previous traditional system.

Experimental Setup

An experimental configuration was established to evaluate the operational performance of the multi-sensor, border security system called Sentinel Eyes in both an indoor/outdoor laboratory setting and an outdoor testing corridor. The experimental configuration utilized a combination of sensors connected to a Raspberry Pi and/or Arduino for real-time processing and evaluation of collected sensor data.

Arduino (the processing unit)

- One or more PIR motion sensors
- One ultrasonic distance sensor
- One magnetic tamper sensor

1. Experimental Hardware Configuration

- The hardware components associated with the experimental configuration are as follows:
- A Raspberry Pi and/or
- One camera module
- One Wi-Fi communication module
- One power supply
- A laptop for viewing the dashboard

All sensor devices were positioned at a height of approximately 1-1.5 meters from the ground to simulate a typical border fence line.

2. Laboratory Testing Area Configuration

The experimental testing area consisted of a length of 5 to 10 meters. Sensors were installed sequentially along this area, with overlapping areas of detection.

The magnetic sensor was mounted to a very small metal framework to replicate a section of fence.

3. Data Collection Methodology

A variety of scenarios were simulated to collect data in order to create and evaluate the datasets:

- Walking/running by a person → labeled Human
- Small animal movement → labeled Animal
- Simulation of bicycle or vehicle passing through → labeled Vehicle
- No wind/no motion → False Alarm
- Physical tampering with the metal frame (shaking/opening) → Tampering

Each scenario resulted in a set of readings from the PIR sensors, the ultrasonic distance sensor, the magnetic sensor, and a digital image from the camera.

4. Real-Time System Testing

The trained machine learning model was deployed on the Raspberry Pi. All incoming data from the various sensors was processed in real-time and classified based on the type of intrusion detected. Alerts were sent when an intrusion occurred. Additionally, a simple graphical interface was developed to display the status of all sensors, alerts, and captured images.

5. Performance Evaluation

The performance of the entire system was evaluated by measuring accuracy, the number of false alarms per hour, the response time to detect an intrusion, and how accurately each event was classified. These results demonstrated that the Sentinel Eyes prototype successfully identified intrusion attempts, significantly reduced false alarms, and provided real-time monitoring capabilities.

Results

The prototype for the sensitial eyes has been tested under several real world conditions,which included human movement ,animal movement,simulated small vehicle movement,and environmental factors(i.e.,wind).All system Successfully Identified and alerted to all types of intrusion and greately reduced false alarams.

The data collected from Experimental trial indicate the accuracy rate for the classification of each type of object or event as follow:

- Human Detection: 92%
- Animal Detection: 88%
- Vehicle Detection: 90%
- False Alarm Detection: 85%

Conculsion

This data illustrates that by combining the use of a multi-sensor array along with an Artificial Intelligence/Machine Learning based classifier, that the Sentinel Eyes is able to provide reliable surveillance capabilities for real time monitoring at borders.

The Sentinel Eyes System demonstrates how the combination of a multi-sensor array with AI/ML algorithms can be used to greatly improve the effectiveness of border surveillance. Through the integration of Passive Infrared (PIR) motion detectors, ultrasonic distance measurement, magnetic tamper detection and camera modules, the system will continue to monitor the area it covers, and provide real-time alerting when an unauthorized person or vehicle enters the monitored area. Furthermore, the machine learning model developed will accurately classify events into one of four categories; human, animal, vehicle, false alarm, and tampering, thereby minimizing unnecessary alerts and increasing the operational accuracy of the system.

Experimental trial results demonstrate a very high level of detection accuracy and low false alarm rates, along with quick response times and the ability to operate in a variety of test environments (indoor and outdoor), thus demonstrating that the prototype version of the Sentinel Eyes is capable of operating reliably in both environments. Additionally, the accuracy versus epoch chart indicates that the model rapidly develops and stabilizes during training. Overall, the Sentinel Eyes presents a low cost, scalable, and low power consumption solution to border security and military zone and remote surveillance applications.

Possible futue enhancements could include the addition of thermal cameras, the integration of drones, the use of more advanced deep learning models, and/or expanding the range of the system. However, even in its prototype form, the Sentinel Eyes demonstrates the potential of intelligent sensor arrays to create smarter and more reliable security systems.

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